

Workshop 2

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1.

- i) $r'^{\wedge}[a-zA-Z_][a-zA-Z_\theta__]*\$'$
- ii) $r'^{\wedge}\backslash d+\$'$
- iii) $r'^{\wedge}\backslash d+\backslash \cdot \backslash d+\$'$
- iv) $r'^{\wedge}''\cdot *?''\$'$
- v) $r'^{\wedge} // \cdot *\$'$
- vi) $r'^{\wedge} / \backslash * \cdot * ? \backslash * / '$
- vii) $r'^{\wedge} \backslash s+ '$
- viii) $r'^{\wedge} (\backslash + \backslash - \backslash * \backslash / \backslash = \backslash ! =) '$
- ix) $r'^{\wedge} \backslash b (\text{if} \mid \text{else} \mid \text{while} \mid \text{return}) \backslash b '$
- x) $r'^{\wedge} \theta x [\theta - q q - F A - F] + \$'$

2. S
9

Program

Statement List

Statement

assignment

Identifier

X

.....

X

Statement List

λ

"=" Expression ";"

Term Expression

Factor

Number

5

.....

5

"+" Term Expression

Factor Term

Number

3

.....

3

"*" Factor Term

Factor Term

Number

2

.....

2

Factor Term

Number

.....

λ

;

b)

S
Program
Statement List
Statement StatementList
IfStatement $\lambda \downarrow$

"if" $\downarrow$ "(" Expression ")" "{" statementList "}" ElsePart
Term Expression
Factor Term $\lambda \downarrow$
Identifier FactorTerm
 $\downarrow$
X
Number $\lambda \downarrow$
 $\downarrow$
0

Error ">" is not Defined

Statement StatementList
Assignment $\lambda \downarrow$
Identifier "=" Expression ";"
 $\downarrow$
y

Term Expression
Factor Term $\lambda \downarrow$
Identifier $\lambda \downarrow$
 $\downarrow$
X

"-" Term Expression
Factor Term $\lambda \downarrow$
Number $\lambda \downarrow$
 $\downarrow$
1

... ElsePart

"else" "{" statementList "}"
statement statementList
assignment $\lambda \downarrow$

Identifier "=" Expression ";"
 $\downarrow$
y

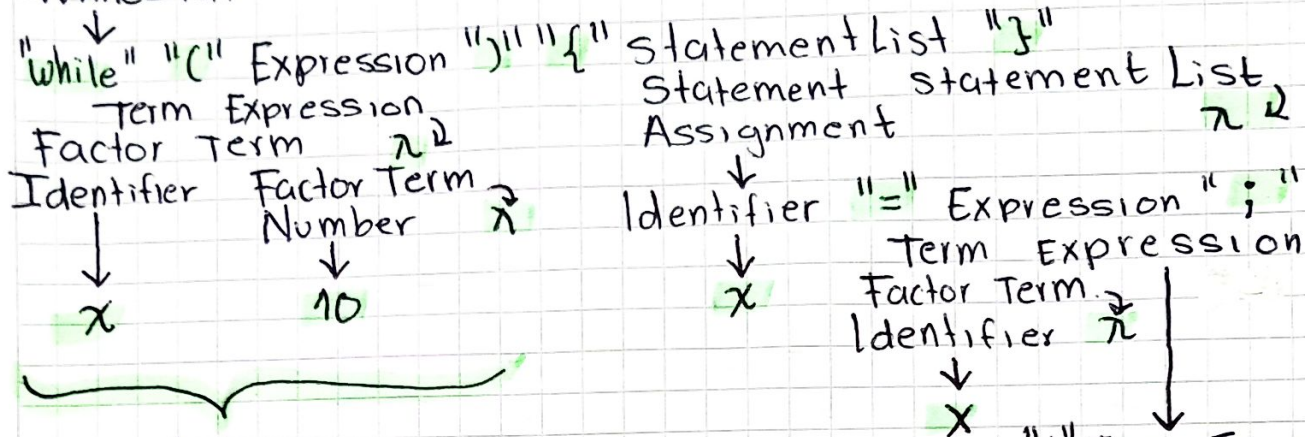
Term Expression
Factor term $\lambda \downarrow$
Number $\lambda \downarrow$
 $\downarrow$
0

```
if (x > 0) {
    y = x - 1;
} else {
    y = 0;
}
```

">" not defined.

C) S

Program
Statement List
Statement StatementList
WhileStatement



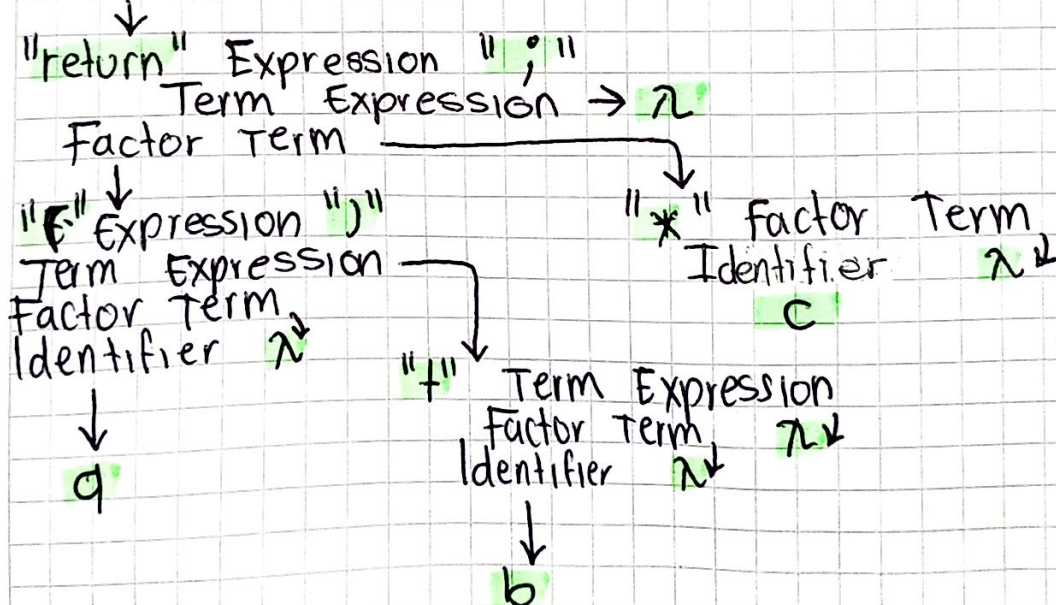
Error "<" is not defined

```
while (x < 10) {
    x = x + 1;
}
```

"<" is not defined.

D) S

Program
Statement List
Statement statementList
ReturnStatement



return (a + b) * c ;